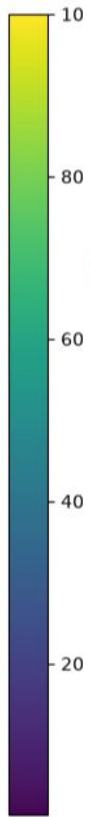

Geospatial Machine Learning II

Édipo Cremon
Rennan Marujo

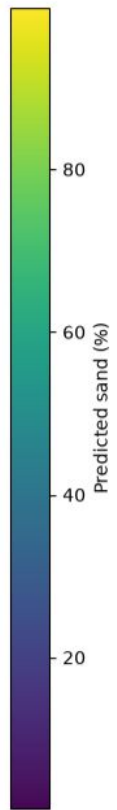
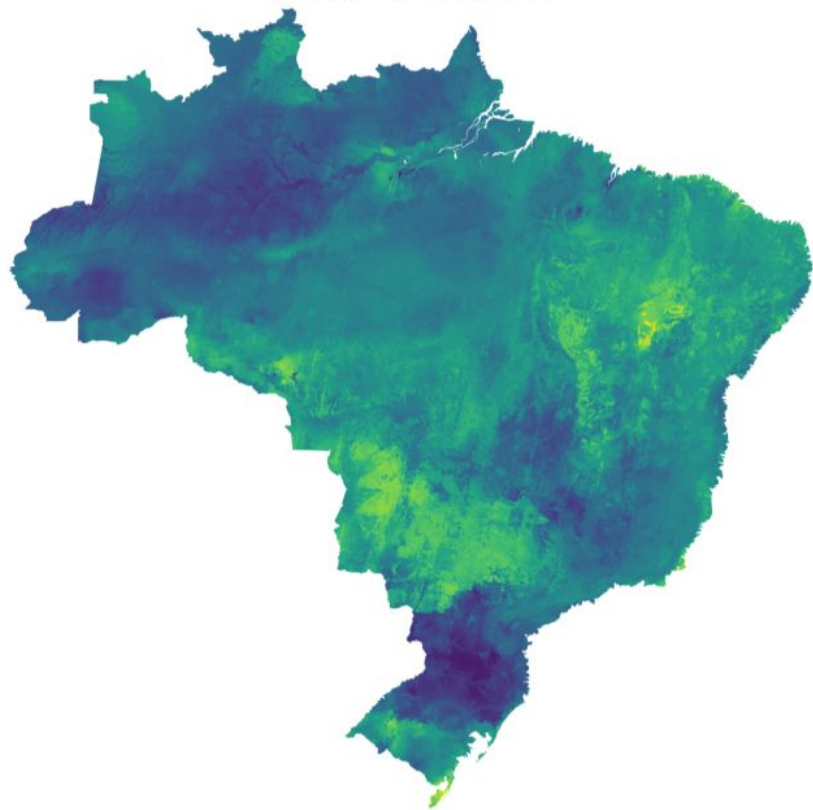


Machine Learning

Random Forest Prediction



Random Forest Prediction





Spatial Regression

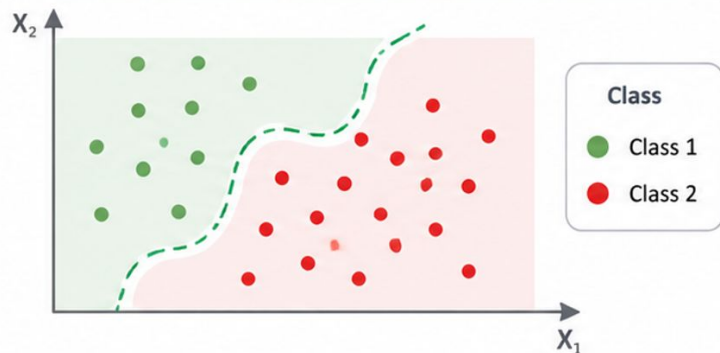
What is the difference?



Classification



Predicts a class or category
(discrete output)



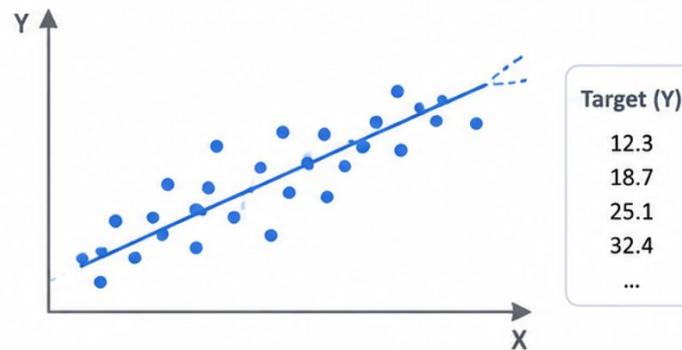
Examples: Spam detection, disease diagnosis, image recognition, sentiment analysis



Regression



Predicts a continuous numerical value
(continuous output)



Examples: House price prediction, sales forecasting, temperature prediction, demand estimation



Key takeaway: Classification answers “which class?” | Regression answers “how much?”

Spatial Regression

Spatial regression incorporates geographic location and spatial dependencies into statistical models

Location and proximity matters!

Traditional regression assumes residuals are randomly distributed and independent

Spatial Regression

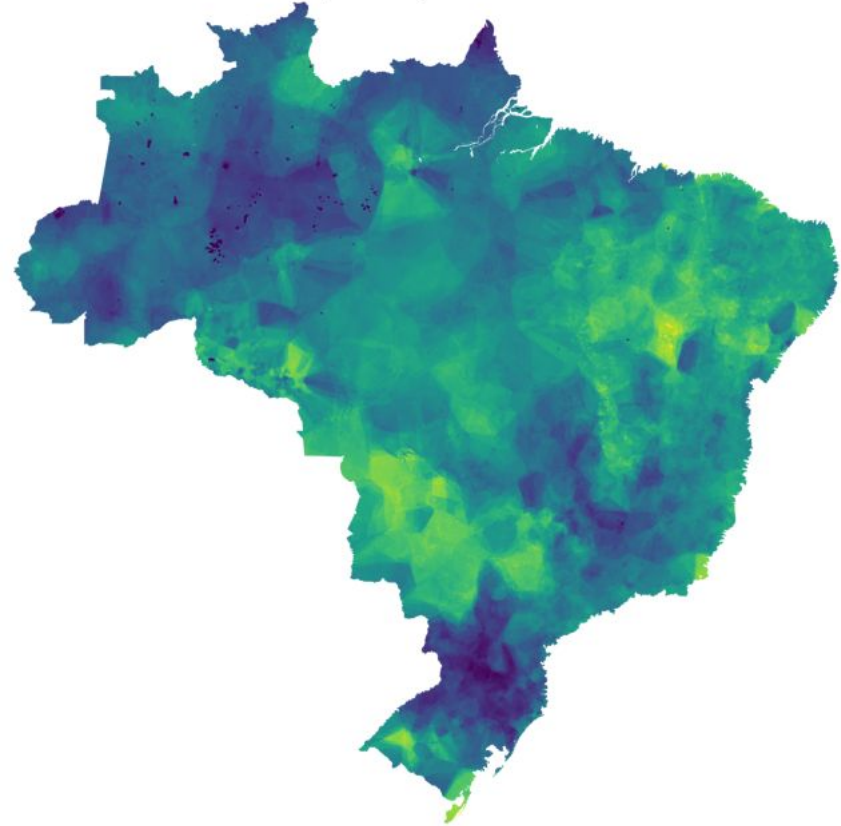
- Spatial lag models `spatialreg::lagsarlm()`
Neighboring values of Y influence Y
- Spatial error models `spatialreg::errorsarlm()`
Spatial dependence remains in the unexplained component
- Geographically weighted regression `spgwr::gwr()`
The relationship between X and Y varies across space

Spatial Regression

Random Forest Prediction



Spatial Lag Prediction



Spatial Machine Learning

1. Classic ML+non-spatial variables+random cross-validation,
2. Classic ML+spatial all variables+random cross-validation,
3. Classic ML+spatial all variables+spatial cross-validation,
4. Classic ML+spatial selected variables+spatial cross-validation,
5. Spatial ML+spatial selected variables+spatial cross-validation.

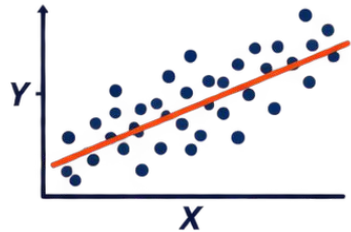
Spatial Machine Learning

1. Classic ML+non-spatial variables+random cross-validation,
2. Classic ML+spatial all variables+random cross-validation,
3. Classic ML+spatial all variables+spatial cross-validation,
4. Classic ML+spatial selected variables+spatial cross-validation,
5. Spatial ML+spatial selected variables+spatial cross-validation.

Spatial Machine Learning

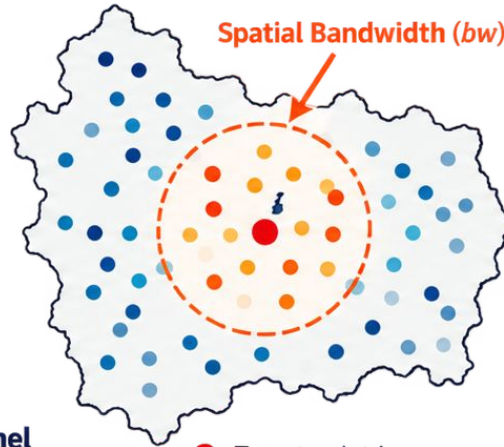
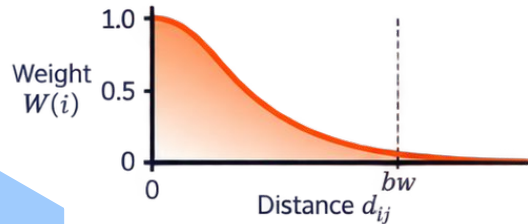
Geographically Weighted Regression (GWR)

Global OLS



A single global regression fits one relationship for the entire study area.

Distance-decay kernel



- Target point i
- Neighbors (higher weight)
- Distant points (lower weight)
- Far points (very low weight)

Local GWR model

$$Y_i = \beta_0(u_i, v_i) + \sum_{k=1}^p \beta_k(u_i, v_i) X_{ki} + \varepsilon_i$$

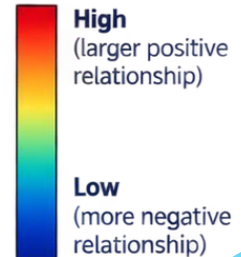
(u_i, v_i) : spatial location of point i

$\beta_k(u_i, v_i)$: location-specific (local) coefficient for predictor k

Local coefficient surface (example: β_1)



Local β_1 value



Spatial Machine Learning

Geographically Weighted Regression (GWR)

GWR addresses spatial heterogeneity more directly than spatial dependence.

Limitations of GWR

- **Limited ability to capture nonlinear relationships**
- **Sensitive to bandwidth and kernel selection**
- **Local multicollinearity may produce unstable coefficients**
- **Risk of overfitting and multiple local inference**
- **Residual spatial autocorrelation may still remain**
- **Local models may generalize poorly to data-sparse or new areas**

Spatial Machine Learning

GWR → local linear models



GRF / GWRF → local Random Forests

GW-SVM → local kernel machines

GWANN → local neural networks

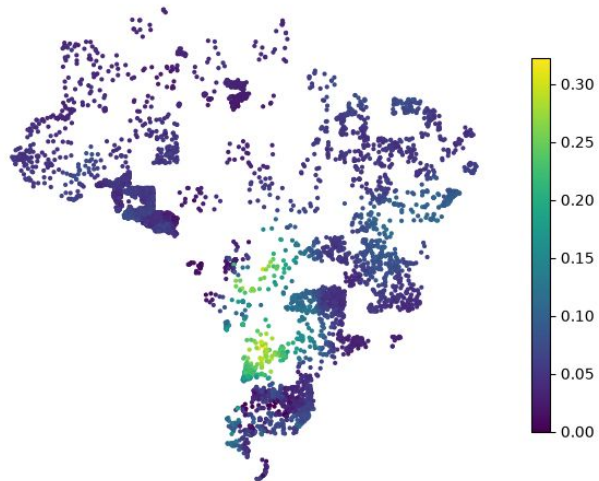
GW-Boosting / GW-XGBoost → local boosted trees

Spatial Random Forest

- Coordinates as predictor variables
- Moran eigenvectors of a connectivity matrix (RF-MEv)
- RF out-of-bag error ordinary kriging (RF-OOB-OK)
- Blended ordinary kriging and RF prediction (OK-RF)
- RF leave-one-out OK (RF-LOO-OK)
- RF spatial interpolation (RFSI)
- RF for spatial prediction (RFsp)
- Generalized-least-squares RF (RF-GLS)
- Exact regionalization RF
- Geographical RF
- Local pixel vector RF
- Space-time RF
- Spatially biased RF

Spatial Random Forest

Local importance of NTSI



Local importance of bio6



Local importance of bio9



Spatial Machine Learning

Communication

A Forest of Forests: A Spatially Weighted and Computationally Efficient Formulation of Geographical Random Forests

Stefanos Georganos ^{1,*} and Stamatis Kalogirou ²

¹ Division of Geoinformatics, KTH Royal Institute of Technology, 10044 Stockholm, Sweden

² Data and Technology for Audit (DATA), European Court of Auditors, 1615 Luxembourg, Luxembourg

* Correspondence: stegeo@kth.se

Abstract: The aim of this paper is to present developments of an advanced geospatial analytics algorithm that improves the prediction power of a random forest regression model while addressing the issue of spatial dependence commonly found in geographical data. We applied the methodology to a simple model of mean household income in the European Union regions to allow easy understanding and reproducibility of the analysis. The results are encouraging and suggest an improvement in the prediction power compared to previous techniques. The algorithm has been implemented in R and is available in the updated version of the SpatialML package in the CRAN repository.

Keywords: spatial machine learning; random forest; spatial heterogeneity; spatial modelling

1. Introduction

Random Forest for Spatial Predictions (RFsp)

PeerJ

Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables

Tomislav Hengl¹, Madlene Nussbaum², Marvin N. Wright³, Gerard B.M. Heuvelink⁴ and Benedikt Gräler⁵

¹ Envirometrix Ltd., Wageningen, Gelderland, Netherlands
² School of Agricultural, Forest and Food Sciences HAFL, Bern University of Applied Sciences BFH, Bern, Switzerland

³ Leibniz Institute for Prevention Research and Epidemiology—BIPS, Bremen, Germany

⁴ Soil Geography and Landscape group, Wageningen Agricultural University, Wageningen, Gelderland, Netherlands

⁵ 52° North Initiative for Geospatial Open Source Software GmbH, Muenster, Germany

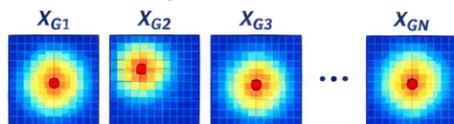
<https://doi.org/10.7717/peerj.5518>

1) Spatial inputs

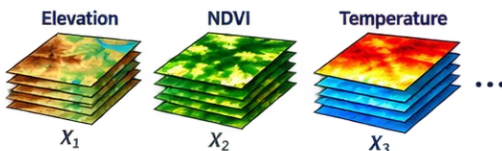
Sampling locations (target variable)



N distance raster layers (Euclidean buffer distances)



Environmental covariate raster layers (X_P , X_R)

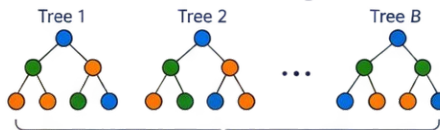


2) Data integration & model

Combined training data matrix

Y (target)	Buffer Distances					Environmental Covariates	
	X_{G1}	X_{G2}	X_{G3}	...	X_{GN}	Elevation	NDVI
Y_1	0.0	1.2	5.6	...	12.3	450	0.32
Y_2	2.1	0.0	3.4	...	8.7	510	0.28
Y_3	4.5	3.2	0.0	...	6.1	620	0.41
...	...	...	...	...	...	...	...
Y_N	7.8	11.4	9.1	...	0.0	380	0.36

Random Forest Engine

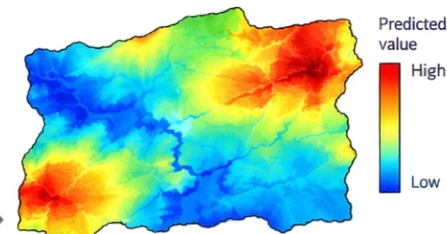


Ensemble of decision trees

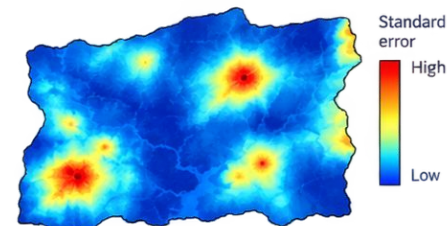
Spatial prediction model (RFsp)

3) Spatial outputs

Predicted target variable (continuous surface)



Prediction uncertainty (standard error)



See: <https://github.com/thengl/geomla>

Geographically Weighted Random Forest (GWRF / GRF)

Communication

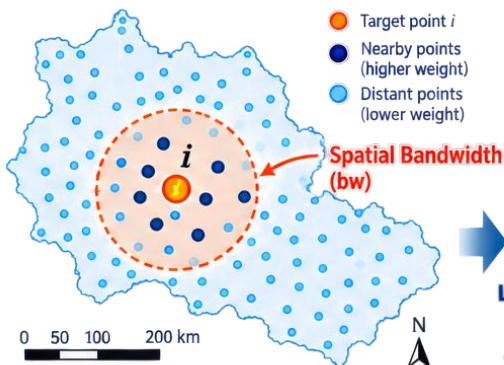
A Forest of Forests: A Spatially Weighted and Computationally Efficient Formulation of Geographical Random Forests

Stefanos Georganos ^{1,*} and Stamatis Kalogirou ²

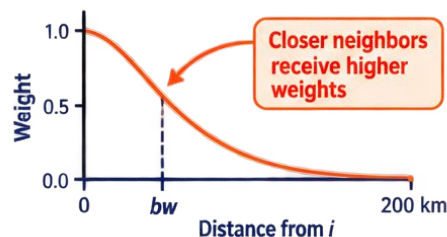
¹ Division of Geoinformatics, KTH Royal Institute of Technology, 10044 Stockholm, Sweden
² Data and Technology for Audit (DATA), European Court of Auditors, 1615 Luxembourg, Luxembourg
* Correspondence: stgeorg@kth.se

1) Spatial neighborhood & kernel

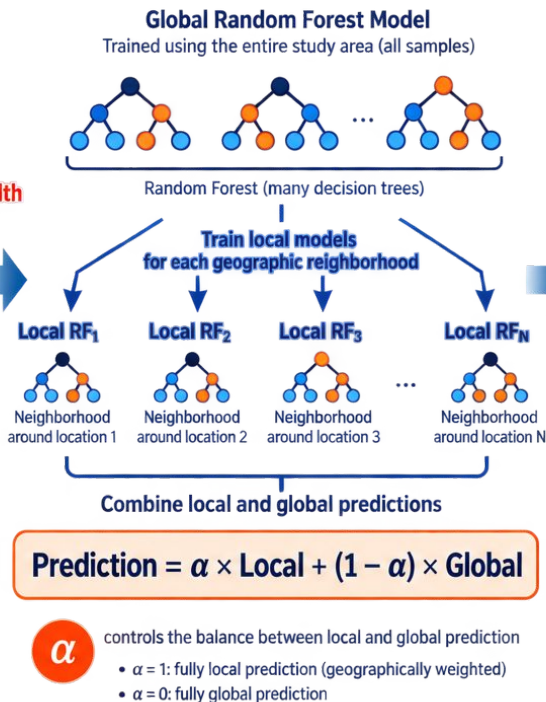
Study area with sample points



Kernel weighting



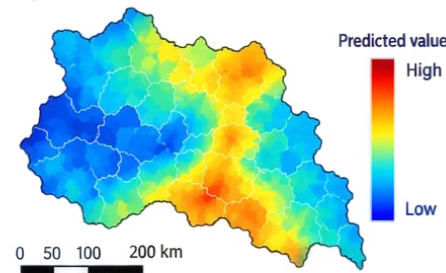
2) Local sub-models fitting



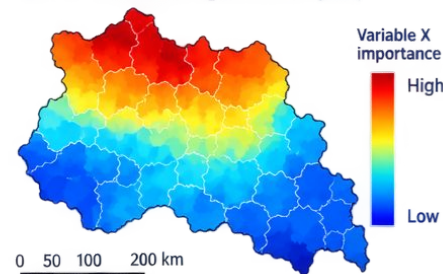
3) Local interpretability & mapping

Spatial Prediction

<https://doi.org/10.3390/ijgi1090471>



Local Variable Importance (LVI)



Model behavior varies across geographic space.

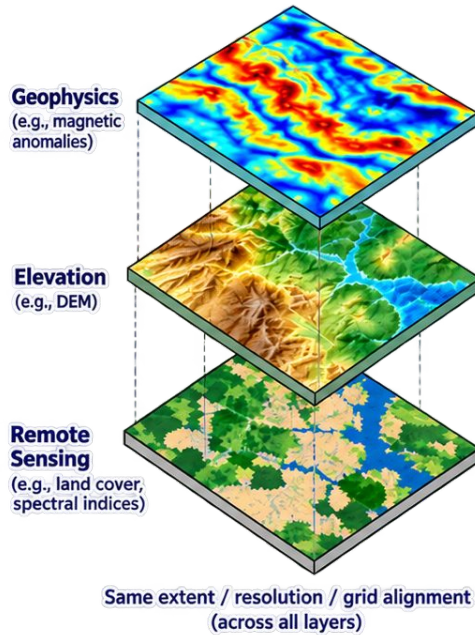
See: <https://cran.r-project.org/web/packages/SpatialML/vignettes/SpatialML.html>
For classification problem, see: <https://github.com/FSantosCodes/GWRFC>

Spatial Random Forest (SRF) - Spatial Context

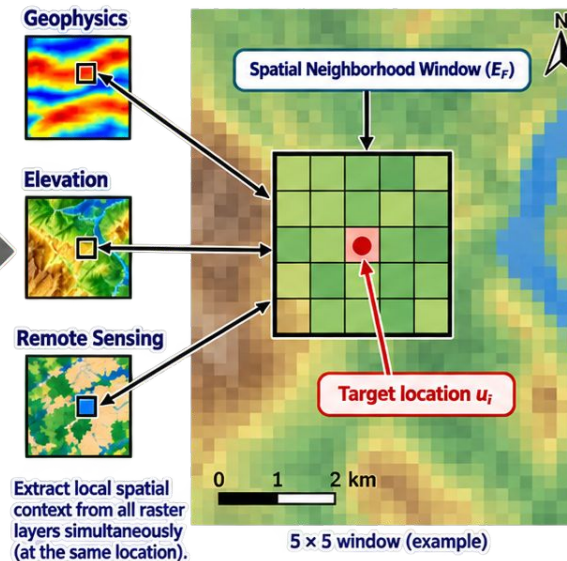
A Truly Spatial Random Forests Algorithm for Geoscience Data Analysis and Modelling

Hassan Talebi^{1,2,4} · Luk J. M. Peeters^{1,3} · Alex Otto² · Raimon Tolosana-Delgado⁵
<https://doi.org/10.1007/s11004-021-09946-w>

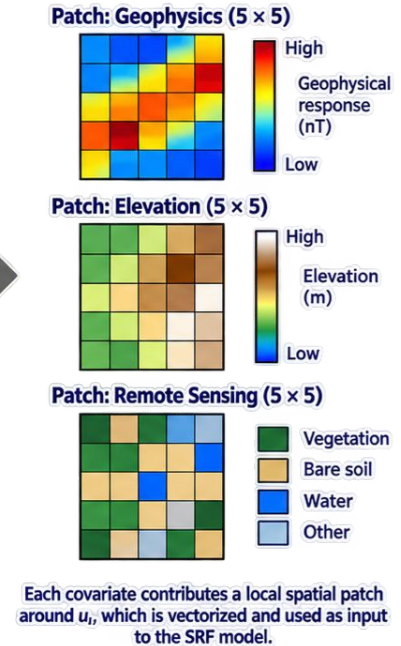
1) Multi-source spatial layers



2) 2D neighborhood window



3) Extrapolated spatial patches



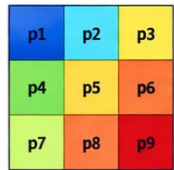
Spatial Random Forest (SRF) - Spatial Context

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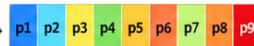
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<https://doi.org/10.1007/s11004-021-09946-w>

1) Patch vectorization and augmentation

Local patch (3 × 3)



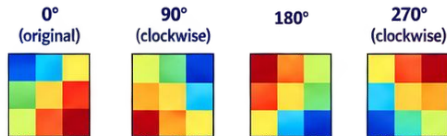
Vectorized patch:
[p₁, p₂, ..., p₉]



Flatten row-by-row
(from top-left to bottom-right)

i Patch values are stored pixel-by-pixel, row-by-row.
Vector: [p₁, p₂, p₃, p₄, p₅, p₆, p₇, p₈, p₉]

Rotation augmentation for orientation invariance



[p₁, p₂, p₃,
p₄, p₅, p₆,
p₇, p₈, p₉]
[p₇, p₄, p₁,
p₈, p₅, p₂,
p₉, p₆, p₃]
[p₉, p₈, p₇,
p₆, p₅, p₄,
p₃, p₂, p₁]
[p₃, p₆, p₉,
p₂, p₅, p₈,
p₁, p₄, p₇]

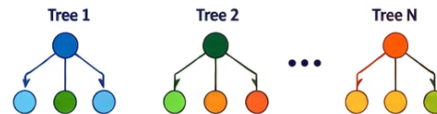
Four augmented versions per patch

2) Dataframe

Sample (ID)	Elevation (Alt)			Geophysics				Remote Sensing				Target Class	
	p1	...	p9	p1	...	p9	p1	...	p9				
1	102	98	...	87	0.21	0.34	...	0.19	0.62	0.55	...	0.48	1
2	95	101	...	83	0.18	0.22	...	0.31	0.58	0.61	...	0.40	0
3	88	90	...	91	0.25	0.28	...	0.33	0.72	0.68	...	0.53	1
4	110	105	...	96	0.19	0.36	...	0.27	0.47	0.51	...	0.46	0
5	93	97	...	89	0.22	0.31	...	0.29	0.66	0.59	...	0.52	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...
N	100	92	...	85	0.20	0.27	...	0.24	0.61	0.57	...	0.49	0

Each row = one vectorized multivariate patch
Columns contain patch pixels from multiple raster layers (covariates).

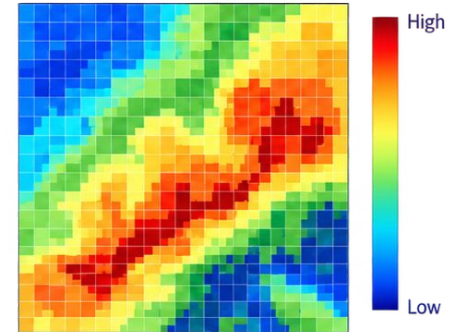
Random Forest Engine



Ensemble prediction (majority vote / probability)

3) Coordinate-free spatial predictions

Spatial prediction map from SRF model



! No absolute (X, Y) coordinates or buffer distances used – enables structural pattern transfer across geographic regions.

Predictions learned from local spatial structure

The model uses local multivariate patch patterns, rather than absolute location, allowing knowledge transfer to new regions with similar spatial structures.



Seminars

Seminars

~20 minutes + questions

Slides until 15/11/2026 (email)

16/11/2026

- Trajectory data structures and functions: e.g. Moving Pandas
- Cloud-optimized rasters: GeoTiff, NetCDF/HDF5, GeoZarr, GRIB3, JPEG2000
- Cloud-optimized vectors: GeoParquet, GeoArrow, FlatGeobuf, shapefile, geopackage
- Geospatial visualization: e.g. *maplibre*, *deck.gl*, Kepler.gl

18/11/2026

- TorchGeo
- PySITS
- Spatial Random Forest – Random Forest Generalized Least Squares (RF-GLS)

Seminars

Seminário 1: Cloud-optimized rasters (GeoTIFF, NetCDF/HDF5, GeoZarr, GRIB3, JPEG2000)

Alunos: Afonso e Giulia

Seminário 2: Spatial Random Forest

Alunos: Amanda e Gustavo

Seminário 3: Cloud-optimized vectors (GeoParquet, GeoArrow, FlatGeobuf, Shapefile, GeoPackage)

Alunos: João e Matheus

Seminário 4: TorchGeo

Alunos: Fawaz e Sandro

Seminário 5: Trajectory data structures and functions (e.g. MovingPandas)

Alunos: Gabrielly e William

Seminário 6: PySITS

Alunos: Lorena e Sofia

Seminário 7: Geospatial visualization (e.g. MapLibre, deck.gl, Kepler.gl)

Aluno: Sérgio

Geospatial Machine Learning II

Édipo Cremon
Rennan Marujo

